0707.4470: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE's — a formula has none of its own. A line's confidence is not a formula's.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0707.4470_F00001	1	1.000	`\nabla \cdot \mathbf{D}=\rho`	$\nabla \cdot \mathbf{D}=\rho$	guishing attributes are that the Gauss constraint $\nabla\cdot\mathbf{D}=\rho$ is exactly conserved in
0707.4470_F00002	2	1.000	`\mathbf{E}=-\nabla \phi`	$\mathbf{E}=-\nabla \phi$	a discrete sense, and electrostatic solutions of the form $\mathbf{E}=-\nabla\phi$ indeed remain
0707.4470_F00003	3	0.677	`\mathbf{B}=\nabla \times \mathbf{A}`	$\mathbf{B}=\nabla \times \mathbf{A}$	sic relation $\mathbf{B}=\nabla\times\mathbf{A}$, where $\mathbf{B}$ is the magnetic flux and $\mathbf{A}$ is the magnetic vector
0707.4470_F00004	3	0.677	`\mathbf{B}`	$\mathbf{B}$	sic relation $\mathbf{B}=\nabla\times\mathbf{A}$, where $\mathbf{B}$ is the magnetic flux and $\mathbf{A}$ is the magnetic vector
0707.4470_F00005	3	1.000	`\nabla \cdot \nabla \times=0`	$\nabla \cdot \nabla \times=0$	potential. Because of the vector calculus identities $\nabla\cdot\nabla\times=0$ and $\nabla\times\nabla=0$, this
0707.4470_F00006	3	1.000	`\nabla \times \nabla=0`	$\nabla \times \nabla=0$	potential. Because of the vector calculus identities $\nabla\cdot\nabla\times=0$ and $\nabla\times\nabla=0$, this
0707.4470_F00007	3	0.959	`\mathbf{A} \mapsto \mathbf{A}+\nabla f`	$\mathbf{A} \mapsto \mathbf{A}+\nabla f$	cally divergence-free. Second, any transformation $\mathbf{A}\mapsto\mathbf{A}+\nabla f$ has no effect on $\mathbf{B}$;
0707.4470_F00008	3	1.000	`\nabla \cdot \mathbf{D}-\rho`	$\nabla \cdot \mathbf{D}-\rho$	is $\nabla\cdot\mathbf{D}-\rho$ (which must be zero by Gauss' law). A similar argument also explains
0707.4470_F00009	3	1.000	`\phi`	ϕ	under constant shifts in the scalar potential ϕ. Therefore, a proper variational
0707.4470_F00010	3	1.000	`\mathrm{d}^{2}=0`	$\mathrm{d}^{2}=0$	tities such as $\mathrm{d}^2=0$ (which encapsulates the previous div-curl-grad relations)
0707.4470_F00011	4	1.000	`F`	F	that are dual to one another (the spacetime forms F and $G=*F$), and hence they
0707.4470_F00012	4	1.000	`G=* F`	$G=* F$	that are dual to one another (the spacetime forms F and $G=*F$), and hence they
0707.4470_F00013	5	0.979	`E`	E	forms, we begin by replacing the electric field with a 1-form E and the magnetic
0707.4470_F00014	5	0.995	`B`	B	flux density by a 2-form B. These have the coordinate expressions
0707.4470_F00015	6	1.000	`\mathbf{E}=\left(E_{x}, E_{y}, E_{z}\right)`	$\mathbf{E}=\left(E_{x}, E_{y}, E_{z}\right)$	where $\mathbf{E}=(E_x,E_y,E_z)$ and $\mathbf{B}=(B_x,B_y,B_z)$. The motivation for choosing E as a
0707.4470_F00016	6	1.000	`\mathbf{B}=\left(B_{x}, B_{y}, B_{z}\right)`	$\mathbf{B}=\left(B_{x}, B_{y}, B_{z}\right)$	where $\mathbf{E}=(E_x,E_y,E_z)$ and $\mathbf{B}=(B_x,B_y,B_z)$. The motivation for choosing E as a
0707.4470_F00017	6	0.995	`\mathbf{E}`	$\mathbf{E}$	where $\mathbf{E}$ is integrated over curves and $\mathbf{B}$ is integrated over surfaces. Similarly,
0707.4470_F00018	6	0.999	`H`	H	1-form H and a 2-form D.
0707.4470_F00019	6	0.999	`D`	D	1-form H and a 2-form D.
0707.4470_F00020	6	1.000	`\epsilon`	ϵ	As shown in Bossavit and Kettunen (2000), we can view ϵ and μ as corresponding
0707.4470_F00021	6	1.000	`\mu`	μ	As shown in Bossavit and Kettunen (2000), we can view ϵ and μ as corresponding
0707.4470_F00022	6	0.968	`*_{\epsilon}`	$*_{\epsilon}$	to Hodge operators $*_\epsilon$ and $*_\mu$, which map the 1-form "fields" to 2-form "fluxes" in
0707.4470_F00023	6	0.968	`*_{\mu}`	$*_{\mu}$	to Hodge operators $*_\epsilon$ and $*_\mu$, which map the 1-form "fields" to 2-form "fluxes" in
0707.4470_F00024	6	1.000	`\epsilon=\epsilon_{0}`	$\epsilon=\epsilon_{0}$	Note that in a vacuum, with $\epsilon=\epsilon_0$ and $\mu=\mu_0$ constant, one can simply express
0707.4470_F00025	6	1.000	`\mu=\mu_{0}`	$\mu=\mu_{0}$	Note that in a vacuum, with $\epsilon=\epsilon_0$ and $\mu=\mu_0$ constant, one can simply express
0707.4470_F00026	6	0.531	`\epsilon_{0}=\mu_{0}=c=1`	$\epsilon_{0}=\mu_{0}=c=1$	that $\epsilon_0=\mu_0=c=1$, and hence ignoring the distinction between $\mathbf{E}$ and $\mathbf{D}$ and
0707.4470_F00027	6	0.925	`\rho \mathrm{d} x \wedge \mathrm{~d} y \wedge \mathrm{~d} z`	$\rho \mathrm{d} x \wedge \mathrm{~d} y \wedge \mathrm{~d} z$	the *charge density* 3*-form* $\rho\ \mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z$, as well as the *current density* 2*-form*
0707.4470_F00028	6	1.000	`J=J_{x} \mathrm{~d} y \wedge \mathrm{~d} z+J_{y} \mathrm{~d} z \wedge \mathrm{~d} x+J_{z} \mathrm{~d} x \wedge \mathrm{~d} y`	$J=J_{x} \mathrm{~d} y \wedge \mathrm{~d} z+J_{y} \mathrm{~d} z \wedge \mathrm{~d} x+J_{z} \mathrm{~d} x \wedge \mathrm{~d} y$	$J=J_x\ \mathrm{d}y\wedge\mathrm{d}z+J_y\ \mathrm{d}z\wedge\mathrm{d}x+J_z\ \mathrm{d}x\wedge\mathrm{d}y$. These are required to satisfy the continuity
0707.4470_F00029	6	1.000	`\partial_{t} \rho+\mathrm{d} J=0`	$\partial_{t} \rho+\mathrm{d} J=0$	of charge condition $\partial_t\rho+\mathrm{d}J=0$, which can be understood as a conservation law
0707.4470_F00030	7	0.992	`\rho`	ρ	2.3. **The Source** 3**-Form.** Likewise, the charge density ρ and current density J
0707.4470_F00031	7	0.992	`J`	J	2.3. **The Source** 3**-Form.** Likewise, the charge density ρ and current density J
0707.4470_F00032	7	1.000	`\mathscr{J}`	$\mathscr{J}$	Having defined $\mathscr{J}$ in this way, the continuity of charge condition simply requires
0707.4470_F00033	7	0.986	`\mathrm{d} \mathscr{J}=0`	$\mathrm{d} \mathscr{J}=0$	that $\mathscr{J}$ be closed, i.e., $\mathrm{d}\mathscr{J}=0$.
0707.4470_F00034	7	1.000	`A`	A	2.4. **Electromagnetic Variational Principle.** Let A be the electromagnetic po-
0707.4470_F00035	7	1.000	`F=\mathrm{d} A`	$F=\mathrm{d} A$	tential 1-form, satisfying $F=\mathrm{d}A$, over the spacetime manifold X. Then define the
0707.4470_F00036	7	1.000	`X`	X	tential 1-form, satisfying $F=\mathrm{d}A$, over the spacetime manifold X. Then define the
0707.4470_F00037	7	1.000	`\alpha`	α	Now, take a variation α of A, where α vanishes on the boundary ∂X. Then the
0707.4470_F00038	7	1.000	`\partial X`	∂X	Now, take a variation α of A, where α vanishes on the boundary ∂X. Then the
0707.4470_F00039	7	1.000	`G=* F=* \mathrm{~d} A`	$G=* F=* \mathrm{~d} A$	2.5. **Variational Derivation of Maxwell's Equations.** Since $G=*F=*\mathrm{d}A$, then

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0707.4470_F00040	7	0.999	`\mathrm{d} G=\mathscr{J}`	$\mathrm{d} G=\mathscr{J}$	clearly Equation 2.1 is equivalent to $\mathrm{d}G = \mathscr{J}$. Furthermore, since $\mathrm{d}^2 = 0$, it follows
0707.4470_F00041	8	1.000	`\mathrm{d} F=\mathrm{d}^{2} A=0`	$\mathrm{d} F=\mathrm{d}^{2} A=0$	that $\mathrm{d}F = \mathrm{d}^2A = 0$. Hence, Maxwell's equations with respect to the Maxwell and
0707.4470_F00042	8	0.991	`(x, y, z, t)`	(x, y, z, t)	Suppose now we choose the standard coordinate system (x, y, z, t) on Minkowski
0707.4470_F00043	8	1.000	`X=\mathbb{R}^{3,1}`	$X=\mathbb{R}^{3,1}$	space $X = \mathbb{R}^{3,1}$, and define E and B through the relation $F = E \wedge \mathrm{d}t + B$. Then a
0707.4470_F00044	8	1.000	`F=E \wedge \mathrm{~d} t+B`	$F=E \wedge \mathrm{~d} t+B$	space $X = \mathbb{R}^{3,1}$, and define E and B through the relation $F = E \wedge \mathrm{d}t + B$. Then a
0707.4470_F00045	8	1.000	`G=* F=H \wedge \mathrm{~d} t-D`	$G=* F=H \wedge \mathrm{~d} t-D$	Likewise, if $G = *F = H \wedge \mathrm{d}t - D$, then Equation 2.3 is equivalent to
0707.4470_F00046	8	1.000	`G`	G	fields, while G acts as a Lagrange multiplier, weakly enforcing the constraint
0707.4470_F00047	8	1.000	`\alpha, \phi, \gamma`	α, ϕ, γ	Then, taking the variation of the action along some α, ϕ, γ (vanishing on ∂X), we
0707.4470_F00048	9	1.000	`\nabla \cdot \mathbf{B}`	$\nabla \cdot \mathbf{B}$	$\nabla \cdot \mathbf{B}$ and $\nabla \cdot \mathbf{D} - \rho$. Therefore, the divergence equations (2.5) and (2.7) can
0707.4470_F00049	9	1.000	`\mathrm{d} A`	$\mathrm{d} A$	exterior derivative $\mathrm{d}A$ of the electromagnetic potential, and not on the value of
0707.4470_F00050	9	1.000	`A \mapsto A+\mathrm{d} f`	$A \mapsto A+\mathrm{d} f$	$A \mapsto A + \mathrm{d}f$ leaves $\mathrm{d}A$, and hence Maxwell's equations, unchanged. Choosing
0707.4470_F00051	9	1.000	`\phi=A(\partial / \partial t)=0`	$\phi=A(\partial / \partial t)=0$	potential $\phi = A(\partial/\partial t) = 0$ (the so-called Weyl gauge or temporal gauge), and
0707.4470_F00052	9	1.000	`\mathrm{d}^{2} A=0`	$\mathrm{d}^{2} A=0$	identity $\mathrm{d}^2A = 0$; they are not actually part of the Euler–Lagrange equations. A
0707.4470_F00053	11	1.000	`n`	n	n-dimensional manifold X is replaced by a discrete mesh—precisely, by a cell
0707.4470_F00054	11	1.000	`K`	K	2-dimensional surface. We will generally denote the complex by K, and a cell in
0707.4470_F00055	11	1.000	`\sigma`	σ	the complex by σ.
0707.4470_F00056	11	0.999	`* K`	$* K$	Given a mesh K, one can construct a *dual mesh* $*K$, where each k-cell σ
0707.4470_F00057	11	0.999	`k`	k	Given a mesh K, one can construct a *dual mesh* $*K$, where each k-cell σ
0707.4470_F00058	11	0.364	`(n-k)`	$(n-k)$	corresponds to a dual $(n-k)$-cell $*\sigma$. ($*K$ is "dual" to K in the sense of a graph
0707.4470_F00059	11	0.364	`* \sigma`	$* \sigma$	corresponds to a dual $(n-k)$-cell $*\sigma$. ($*K$ is "dual" to K in the sense of a graph
0707.4470_F00060	11	0.364	`(* K`	$(* K$	corresponds to a dual $(n-k)$-cell $*\sigma$. ($*K$ is "dual" to K in the sense of a graph
0707.4470_F00061	11	0.972	`(n-1)`	$(n-1)$	corresponding n-simplices share an $(n-1)$-simplex, and so on. This is called
0707.4470_F00062	11	1.000	`\alpha^{k}`	α^{k}	ferential forms. A discrete k-form α^k assigns a real number to each oriented
0707.4470_F00063	11	1.000	`\sigma^{k}`	σ^{k}	k-dimensional cell σ^k in the mesh K. (The superscripts k are not actually re-
0707.4470_F00064	11	1.000	`\left\langle\alpha^{k}, \sigma^{k}\right\rangle`	$\left\langle\alpha^{k}, \sigma^{k}\right\rangle$	form or cell we are dealing with.) This value is denoted by $\left\langle \alpha^k, \sigma^k \right\rangle$, and can be
0707.4470_F00065	12	1.000	`\sigma^{1}`	σ^{1}	show one particular primal edge σ^1 (left) and its corresponding dual edge $*\sigma^1$
0707.4470_F00066	12	1.000	`* \sigma^{1}`	$* \sigma^{1}$	show one particular primal edge σ^1 (left) and its corresponding dual edge $*\sigma^1$
0707.4470_F00067	12	1.000	`\mathrm{CH}\left(\sigma^{1}, * \sigma^{1}\right)`	$\mathrm{CH}\left(\sigma^{1}, * \sigma^{1}\right)$	(right); the convex hull of these cells $\mathrm{CH}(\sigma^1, *\sigma^1)$ is shaded dark grey.
0707.4470_F00068	12	1.000	`\langle\cdot, \cdot\rangle`	$\langle\cdot, \cdot\rangle$	are called *chains*, and discrete differential forms are *cochains*, where $\langle\cdot,\cdot\rangle$ is the
0707.4470_F00069	12	0.994	`n-k`	$n-k$	a natural correspondence between primal k-forms and dual $(n-k)$-forms, since
0707.4470_F00070	12	0.884	`k+1`	$k+1$	Therefore, if α is a discrete differential k-form, then the $(k+1)$-form $\mathrm{d}\alpha$ is defined
0707.4470_F00071	12	0.884	`\mathrm{d} \alpha`	$\mathrm{d} \alpha$	Therefore, if α is a discrete differential k-form, then the $(k+1)$-form $\mathrm{d}\alpha$ is defined
0707.4470_F00072	12	0.604	`(k+1)`	$(k+1)$	on any $(k+1)$-chain σ by
0707.4470_F00073	12	1.000	`\partial \sigma`	$\partial \sigma$	where $\partial\sigma$ is the k-chain boundary of σ. For this reason, d is often called the
0707.4470_F00074	13	1.000	`* \alpha`	$* \alpha$	versions can be substituted. Given a discrete form α, its *Hodge star* $*\alpha$ is defined
0707.4470_F00075	13	1.000	`|\sigma|`	$\|\sigma\|$	where $\|\sigma\|$ and $\|*\sigma\|$ are the volumes of these elements, and κ is the causality
0707.4470_F00076	13	1.000	`|* \sigma|`	$\|* \sigma\|$	where $\|\sigma\|$ and $\|*\sigma\|$ are the volumes of these elements, and κ is the causality
0707.4470_F00077	13	1.000	`\kappa`	κ	where $\|\sigma\|$ and $\|*\sigma\|$ are the volumes of these elements, and κ is the causality
0707.4470_F00078	13	0.755	`\mathbb{L}^{2}`	$\mathbb{L}^{2}$	*Inner Product.* Define the $\mathbb{L}^2$ *inner product* $(\cdot,\cdot)$ between two primal k-forms to
0707.4470_F00079	13	0.755	`(\cdot, \cdot)`	$(\cdot, \cdot)$	*Inner Product.* Define the $\mathbb{L}^2$ *inner product* $(\cdot,\cdot)$ between two primal k-forms to
0707.4470_F00080	13	0.978	`\mathrm{CH}(\sigma, * \sigma)`	$\mathrm{CH}(\sigma, * \sigma)$	where the sum is taken over all k-dimensional elements σ, and $\mathrm{CH}(\sigma, *\sigma)$ is
0707.4470_F00081	13	1.000	`\sigma \cup * \sigma`	$\sigma \cup * \sigma$	the n-dimensional convex hull of $\sigma \cup *\sigma$ (see Figure 1). The final equality holds
0707.4470_F00082	13	0.991	`|\mathrm{CH}(\sigma, * \sigma)|=\binom{n}{k}^{-1}| \sigma||* \sigma|`	$\|\mathrm{CH}(\sigma, * \sigma)\|=\binom{n}{k}^{-1}\| \sigma\|\|* \sigma\|$	one another, and hence $\|\mathrm{CH}(\sigma, *\sigma)\| = \binom{n}{k}^{-1} \|\sigma\|\|*\sigma\|$. (Indeed, this is one of the
0707.4470_F00083	13	1.000	`\alpha \wedge * \beta`	$\alpha \wedge * \beta$	primal-dual convex hulls.) This inner product can be expressed in terms of $\alpha \wedge *\beta$,
0707.4470_F00084	13	0.865	`\delta`	δ	d and $*$, we immediately have a discrete codifferential δ, with the same formal
0707.4470_F00085	13	0.995	`\mathrm{d}, *, \delta`	$\mathrm{d}, *, \delta$	dual discrete forms, along with the corresponding operators $\mathrm{d}, *, \delta$, for the case
0707.4470_F00086	13	1.000	`\sigma_{i}^{k}`	σ_{i}^{k}	on each k-cell of the mesh. That is, given a list of k-cells σ_i^k, the entries of the

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0707.4470_F00087	13	0.803	`\alpha_{i}=\left\langle\alpha, \sigma_{i}^{k}\right \rangle`	$\alpha_{i}=\left\langle\alpha, \sigma_{i}^{k}\right\rangle$	vector are $\alpha_i = \left\langle \alpha, \sigma_i^k \right\rangle$. The exterior derivative d, taking k-forms to $(k+1)$-forms,
0707.4470_F00088	14	1.000	`k=0,1,2,3`	$k=0,1,2,3$	cells of the primal mesh, for $k=0,1,2,3$; the bottom row shows the location of the
0707.4470_F00089	14	0.965	`*`	$*$	and δ map "horizontally" between k and $(k+1)$ forms, while the Hodge star $*$
0707.4470_F00090	14	1.000	`*^{-1}`	$*^{-1}$	and its inverse $*^{-1}$ map "vertically" between primal and dual forms.
0707.4470_F00091	14	0.510	`\kappa\left(\sigma_{i}^{k}\right) \frac{\left|* \sigma_{i}^{k}\right|}{\left|\sigma_{i}^{k}\right|}`	$\kappa\left(\sigma_{i}^{k}\right) \frac{\left\lvert * \sigma_{i}^{k}\right\rvert}{\left\lvert\sigma_{i}^{k}\right\rvert}$	of the diagonal Hodge star, it is the diagonal matrix with entries $\kappa\left(\sigma_i^k\right)\frac{\lvert *\sigma_i^k\rvert}{\lvert\sigma_i^k\rvert}$. The
0707.4470_F00092	14	1.000	`\partial K`	∂K	∂K. For example, in electromagnetism, we may wish to set initial conditions for
0707.4470_F00093	14	1.000	`t_{0}`	t_{0}	E and B at time t_0 — but while B is defined on ∂K at t_0, E is not. In fact, as we
0707.4470_F00094	14	1.000	`t_{0}, E`	t_{0}, E	E and B at time t_0 — but while B is defined on ∂K at t_0, E is not. In fact, as we
0707.4470_F00095	14	1.000	`t_{1}`	t_{1}	will see, E lives on edges that are extruded between the time slices t_0 and t_1, so
0707.4470_F00096	14	1.000	`t_{1 / 2}`	$t_{1 / 2}$	unless we modify our definitions, we can only initialize E at the half-step $t_{1/2}$.
0707.4470_F00097	15	0.608	`n-1`	$n-1$	addition of a dual vertex at the circumcenter of each boundary $(n-1)$-simplex, in
0707.4470_F00098	15	0.844	`(k-1)`	$(k-1)$	dimensional shell. That is, each boundary $(k-1)$-simplex can be thought of
0707.4470_F00099	15	0.997	`*(\partial K)`	$*(\partial K)$	$*(\partial K)$. In fact, the boundary of the dual is now equal to the dual of the boundary,
0707.4470_F00100	15	1.000	`\partial(* K)=*(\partial K)`	$\partial(* K)=*(\partial K)$	i.e., $\partial(*K)=*(\partial K)$. Returning to the example of initial conditions on E and B, we
0707.4470_F00101	15	1.000	`t_{-\epsilon}`	$t_{-\epsilon}$	vanishingly thin faces (i.e., extruded between some $t_{-\epsilon}$ and t_0 for $\epsilon \to 0$). Notice
0707.4470_F00102	15	1.000	`\epsilon \rightarrow 0`	$\epsilon \rightarrow 0$	vanishingly thin faces (i.e., extruded between some $t_{-\epsilon}$ and t_0 for $\epsilon \to 0$). Notice
0707.4470_F00103	16	0.862	`k-1`	$k-1$	boundary terms. If α is a primal $(k-1)$-form and β is a primal k-form, then
0707.4470_F00104	16	0.862	`\beta`	β	boundary terms. If α is a primal $(k-1)$-form and β is a primal k-form, then
0707.4470_F00105	16	0.894	`* \beta`	$* \beta$	In the boundary integral, α is still a primal $(k-1)$-form on ∂K, while $*\beta$ is an
0707.4470_F00106	16	0.991	`F, G`	F, G	First, we will find a sensible way to define the discrete forms F, G, and $\mathscr{J}$ on a
0707.4470_F00107	16	0.989	`\mathbb{R}^{3,1}`	$\mathbb{R}^{3,1}$	generalized spacetime meshes, e.g., an arbitrary meshing of $\mathbb{R}^{3,1}$ by 4-simplices.
0707.4470_F00108	16	1.000	`p`	p	more abstract perspective of higher-level "p-form" versions of electromagnetism
0707.4470_F00109	16	1.000	`\Delta x, \Delta y, \Delta z, \Delta t`	$\Delta x, \Delta y, \Delta z, \Delta t$	let us also suppose that the grid has uniform space and time steps $\Delta x, \Delta y, \Delta z, \Delta t$.
0707.4470_F00110	17	0.925	`E_{x} \Delta x \Delta t`	$E_{x} \Delta x \Delta t$	*six types of 2-faces in a 4-D rectangular grid.* Simply assign the values $E_x \Delta x \Delta t$
0707.4470_F00111	17	1.000	`x t`	$x t$	to faces parallel to the xt-plane, $E_y \Delta y \Delta t$ to faces parallel to the yt-plane, and
0707.4470_F00112	17	1.000	`E_{y} \Delta y \Delta t`	$E_{y} \Delta y \Delta t$	to faces parallel to the xt-plane, $E_y \Delta y \Delta t$ to faces parallel to the yt-plane, and
0707.4470_F00113	17	1.000	`y t`	$y t$	to faces parallel to the xt-plane, $E_y \Delta y \Delta t$ to faces parallel to the yt-plane, and
0707.4470_F00114	17	1.000	`E_{z} \Delta z \Delta t`	$E_{z} \Delta z \Delta t$	$E_z \Delta z \Delta t$ to faces parallel to the zt-plane. Likewise, assign $B_x \Delta y \Delta z$ to faces
0707.4470_F00115	17	1.000	`z t`	$z t$	$E_z \Delta z \Delta t$ to faces parallel to the zt-plane. Likewise, assign $B_x \Delta y \Delta z$ to faces
0707.4470_F00116	17	1.000	`B_{x} \Delta y \Delta z`	$B_{x} \Delta y \Delta z$	$E_z \Delta z \Delta t$ to faces parallel to the zt-plane. Likewise, assign $B_x \Delta y \Delta z$ to faces
0707.4470_F00117	17	1.000	`y z`	$y z$	parallel to the yz-plane, $B_y \Delta z \Delta x$ to faces parallel to the xz-plane, and $B_z \Delta x \Delta y$
0707.4470_F00118	17	1.000	`B_{y} \Delta z \Delta x`	$B_{y} \Delta z \Delta x$	parallel to the yz-plane, $B_y \Delta z \Delta x$ to faces parallel to the xz-plane, and $B_z \Delta x \Delta y$
0707.4470_F00119	17	1.000	`x z`	$x z$	parallel to the yz-plane, $B_y \Delta z \Delta x$ to faces parallel to the xz-plane, and $B_z \Delta x \Delta y$
0707.4470_F00120	17	1.000	`B_{z} \Delta x \Delta y`	$B_{z} \Delta x \Delta y$	parallel to the yz-plane, $B_y \Delta z \Delta x$ to faces parallel to the xz-plane, and $B_z \Delta x \Delta y$
0707.4470_F00121	17	0.998	`x y`	$x y$	to faces parallel to the xy-plane. This is pictured in Figure 4.
0707.4470_F00122	17	0.930	`\left[x_{k}, x_{k+1}\right] \times`	$\left[x_{k}, x_{k+1}\right] \times$	Let us look at these values on the faces of a typical 4-rectangle $[x_k, x_{k+1}] \times$
0707.4470_F00123	17	1.000	`\left[y_{l}, y_{l+1}\right] \times\left[z_{m}, z_{m+1}\right] \times\left[t_{n}, t_{n+1}\right]`	$\left[y_{l}, y_{l+1}\right] \times\left[z_{m}, z_{m+1}\right] \times\left[t_{n}, t_{n+1}\right]$	$[y_l, y_{l+1}] \times [z_m, z_{m+1}] \times [t_n, t_{n+1}]$. To simplify the notation, we can index each
0707.4470_F00124	17	0.965	`\left.F\right|_{k+\frac{1}{2}, l, m} ^{n+\frac{1}{2}}`	$\left.F\right\rvert_{k+\frac{1}{2}, l, m}^{n+\frac{1}{2}}$	value of F by the midpoint of the 2-face on which it lives: for example, $F\rvert_{k+\frac{1}{2},l,m}^{n+\frac{1}{2}}$
0707.4470_F00125	17	1.000	`\left[x_{k}, x_{k+1}\right] \times\left\{y_{l}\right\} \times\left\{z_{m}\right\} \times\left[t_{n}, t_{n+1}\right]`	$\left[x_{k}, x_{k+1}\right] \times\left\{y_{l}\right\} \times\left\{z_{m}\right\} \times\left[t_{n}, t_{n+1}\right]$	is stored on the face $[x_k, x_{k+1}] \times \{y_l\} \times \{z_m\} \times [t_n, t_{n+1}]$, parallel to the xt-plane.
0707.4470_F00126	18	1.000	`\mathrm{d} F`	$\mathrm{d} F$	at the DEC interpretation of $\mathrm{d}F$. Since $\mathrm{d}F$ is a discrete 3-form, it takes values on
0707.4470_F00127	20	1.000	`\mathbf{D}=\epsilon \mathbf{E}, \mathbf{B}=\mu \mathbf{H}`	$\mathbf{D}=\epsilon \mathbf{E}, \mathbf{B}=\mu \mathbf{H}$	tion 5.2 for details) and making the substitutions $\mathbf{D}=\epsilon\mathbf{E}, \mathbf{B}=\mu\mathbf{H}$, the remaining
0707.4470_F00128	21	1.000	`\kappa=1`	$\kappa=1$	the value $\kappa=1$. Second, there are rectangular faces that live between time steps.
0707.4470_F00129	21	1.000	`\kappa=-1`	$\kappa=-1$	they have one timelike edge, these faces satisfy $\kappa=-1$. Again, the circumcentric-

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0707.4470_F00130	21	1.000	`E \Delta t`	$E \Delta t$	Therefore, let us assign B to the purely spacelike faces and $E\Delta t$ to the mixed
0707.4470_F00131	21	1.000	`H \Delta t`	$H \Delta t$	$H\Delta t$ and spacelike dual faces should store D; see Figure 6.
0707.4470_F00132	21	1.000	`B^{n}`	B^{n}	Section 3.2. This leads to the arrays B^n and H^n at whole time steps n, and $E^{n+1/2}$
0707.4470_F00133	21	1.000	`H^{n}`	H^{n}	Section 3.2. This leads to the arrays B^n and H^n at whole time steps n, and $E^{n+1/2}$
0707.4470_F00134	21	1.000	`E^{n+1 / 2}`	$E^{n+1 / 2}$	Section 3.2. This leads to the arrays B^n and H^n at whole time steps n, and $E^{n+1/2}$
0707.4470_F00135	21	1.000	`D^{n+1 / 2}`	$D^{n+1 / 2}$	and $D^{n+1/2}$ at half time steps. Let d_1 denote the edges-to-faces incidence matrix
0707.4470_F00136	21	1.000	`\mathrm{d}_{1}`	d_{1}	and $D^{n+1/2}$ at half time steps. Let d_1 denote the edges-to-faces incidence matrix
0707.4470_F00137	21	1.000	`\mathrm{d}_{1}^{T}`	d_{1}^{T}	Similarly, the transpose d_1^T corresponds to the exterior derivative from spatial
0707.4470_F00138	21	1.000	`\mathrm{d} F=0`	$\mathrm{d} F=0$	dual 1-forms to dual 2-forms. Then the equation $\mathrm{d}F=0$, evaluated on all prismal
0707.4470_F00139	23	1.000	`\Delta t_{\sigma}=t_{\sigma}^{n+1}-t_{\sigma}^{n}`	$\Delta t_{\sigma}=t_{\sigma}^{n+1}-t_{\sigma}^{n}$	For example, one might assign each face a fixed time step size $\Delta t_\sigma = t_\sigma^{n+1} - t_\sigma^n$,
0707.4470_F00140	23	1.000	`\Delta t`	Δt	taking equal time steps *within* each element, but with Δt varying *across* elements.
0707.4470_F00141	23	1.000	`\Theta_{\sigma} \cap \Theta_{\sigma^{\prime}}=\left \{t_{0}\right\}`	$\Theta_{\sigma} \cap \Theta_{\sigma^{\prime}}=\left\{t_{0}\right\}$	no two faces take the same time step: that is, $\Theta_\sigma \cap \Theta_{\sigma'} = \{t_0\}$ for $\sigma \neq \sigma'$.
0707.4470_F00142	23	1.000	`\sigma \neq \sigma^{\prime}`	$\sigma \neq \sigma^{\prime}$	no two faces take the same time step: that is, $\Theta_\sigma \cap \Theta_{\sigma'} = \{t_0\}$ for $\sigma \neq \sigma'$.
0707.4470_F00143	23	1.000	`e`	e	In order to keep proper time at the edges e where multiple faces with different
0707.4470_F00144	23	1.000	`t_{e}^{k+1 / 2}=\left(t_{e}^{k+1}+t_{e}^{k}\right) / 2`	$t_{e}^{k+1 / 2}=\left(t_{e}^{k+1}+t_{e}^{k}\right) / 2$	where $t_e^{k+1/2} = (t_e^{k+1} + t_e^k)/2$. The values stored on a primal AVI mesh are shown
0707.4470_F00145	23	1.000	`\Theta_{e} \supset \Theta_{\sigma}`	$\Theta_{e} \supset \Theta_{\sigma}$	Since $\Theta_e \supset \Theta_\sigma$ when $e \subset \sigma$, each spatial edge e takes more time steps than any
0707.4470_F00146	23	1.000	`e \subset \sigma`	$e \subset \sigma$	Since $\Theta_e \supset \Theta_\sigma$ when $e \subset \sigma$, each spatial edge e takes more time steps than any
0707.4470_F00147	24	1.000	`\nVdash_{\left\{t_{\sigma}^{n}=t_{e}^{m}\right\}}`	$\nVdash_{\left\{t_{\sigma}^{n}=t_{e}^{m}\right\}}$	where $\nVdash_{\{t_\sigma^n = t_e^m\}}$ equals 1 when face σ has $t_\sigma^n = t_e^m$ for some n, and 0 otherwise.
0707.4470_F00148	24	1.000	`t_{\sigma}^{n}=t_{e}^{m}`	$t_{\sigma}^{n}=t_{e}^{m}$	where $\nVdash_{\{t_\sigma^n = t_e^m\}}$ equals 1 when face σ has $t_\sigma^n = t_e^m$ for some n, and 0 otherwise.
0707.4470_F00149	24	1.000	`t_{\sigma}^{n+1}`	t_{σ}^{n+1}	(1) Pick the minimum time t_σ^{n+1} where B_σ^{n+1} has not yet been computed.
0707.4470_F00150	24	1.000	`B_{\sigma}^{n+1}`	B_{σ}^{n+1}	(1) Pick the minimum time t_σ^{n+1} where B_σ^{n+1} has not yet been computed.
0707.4470_F00151	24	1.000	`H_{\sigma}^{n+1}=*_{\mu}^{-1} B_{\sigma}^{n+1}`	$H_{\sigma}^{n+1}=*_{\mu}^{-1} B_{\sigma}^{n+1}$	(3) Update $H_\sigma^{n+1} = *_\mu^{-1} B_\sigma^{n+1}$.
0707.4470_F00152	24	1.000	`D_{e}^{m+3 / 2}`	$D_{e}^{m+3 / 2}$	(4) Advance $D_e^{m+3/2}$ on neighboring edges $e \subset \sigma$ according to Equation 4.4.
0707.4470_F00153	24	1.000	`E_{e}^{m+3 / 2}=*_{\epsilon}^{-1} D_{e}^{m+3 / 2}`	$E_{e}^{m+3 / 2}=*_{\epsilon}^{-1} D_{e}^{m+3 / 2}$	(5) Update $E_e^{m+3/2} = *_\epsilon^{-1} D_e^{m+3/2}$.
0707.4470_F00154	24	0.902	`\mathbf{x}`	$\mathbf{x}$	pared to the AVI for elasticity, A plays the role of the positions $\mathbf{x}$, while E plays
0707.4470_F00155	24	1.000	`\dot{\mathbf{x}}`	$\dot{\mathbf{x}}$	the role of the (negative) velocities $\dot{\mathbf{x}}$. The algorithm is given as pseudocode in
0707.4470_F00156	25	0.748	`A_{e} \leftarrow A_{e}^{0}, E_{e} \leftarrow E_{e}^{1 / 2}, \tau_{e} \leftarrow t_{0} / /`	$A_{e} \leftarrow A_{e}^{0}, E_{e} \leftarrow E_{e}^{1 / 2}, \tau_{e} \leftarrow t_{0} / /$	$A_e \leftarrow A_e^0$, $E_e \leftarrow E_e^{1/2}$, $\tau_e \leftarrow t_0$ // *Store initial field values and times*
0707.4470_F00157	25	1.000	`\tau_{\sigma} \leftarrow t_{0}`	$\tau_{\sigma} \leftarrow t_{0}$	$\tau_\sigma \leftarrow t_0$
0707.4470_F00158	25	0.999	`t_{\sigma}^{1}`	t_{σ}^{1}	Compute the next update time t_σ^1
0707.4470_F00159	25	0.851	`Q . \mathrm{push}\left(t_{\sigma}^{1}, \sigma\right) / /`	$Q . \mathrm{push}\left(t_{\sigma}^{1}, \sigma\right) / /$	Q.push(t_σ^1, σ) // *Push element onto queue with its next update time*
0707.4470_F00160	25	0.873	`(t, \sigma) \leftarrow Q . \operatorname{pop}() / /`	$(t, \sigma) \leftarrow Q . \operatorname{pop}() / /$	$(t, \sigma) \leftarrow Q$.pop() // *Pop next element σ and time t from queue*
0707.4470_F00161	25	0.873	`t`	t	$(t, \sigma) \leftarrow Q$.pop() // *Pop next element σ and time t from queue*
0707.4470_F00162	25	1.000	`A_{e} \leftarrow A_{e}-E_{e}\left(t-\tau_{e}\right) / /`	$A_{e} \leftarrow A_{e}-E_{e}\left(t-\tau_{e}\right) / /$	$A_e \leftarrow A_e - E_e(t - \tau_e)$ // *Update neighboring values of A at time t*
0707.4470_F00163	25	0.570	`t<`	$t <$	**if** $t <$ final-time **then**
0707.4470_F00164	25	1.000	`B_{\sigma} \leftarrow \mathrm{d}_{1} A_{e}`	$B_{\sigma} \leftarrow \mathrm{d}_{1} A_{e}$	$B_\sigma \leftarrow \mathrm{d}_1 A_e$
0707.4470_F00165	25	1.000	`H_{\sigma} \leftarrow *_{\mu} B_{\sigma}`	$H_{\sigma} \leftarrow *_{\mu} B_{\sigma}$	$H_\sigma \leftarrow *_\mu B_\sigma$
0707.4470_F00166	25	0.998	`D_{e} \leftarrow *_{\epsilon} E_{e}`	$D_{e} \leftarrow *_{\epsilon} E_{e}$	$D_e \leftarrow *_\epsilon E_e$
0707.4470_F00167	25	1.000	`D_{e} \leftarrow D_{e}+\mathrm{d}_{1}(e, \sigma) H_{\sigma}\left(t-\tau_{\sigma}\right)`	$D_{e} \leftarrow D_{e}+\mathrm{d}_{1}(e, \sigma) H_{\sigma}\left(t-\tau_{\sigma}\right)$	$D_e \leftarrow D_e + \mathrm{d}_1(e, \sigma) H_\sigma (t - \tau_\sigma)$
0707.4470_F00168	25	1.000	`E_{e} \leftarrow *_{\epsilon} D_{e}`	$E_{e} \leftarrow *_{\epsilon} D_{e}$	$E_e \leftarrow *_\epsilon D_e$
0707.4470_F00169	25	0.999	`\tau_{\sigma} \leftarrow t / /`	$\tau_{\sigma} \leftarrow t / /$	$\tau_\sigma \leftarrow t$ // *Update element's time*
0707.4470_F00170	25	0.997	`t_{\sigma}^{\text {next }}`	$t_{\sigma}^{\text {next }}$	Compute the next update time t_σ^{next}
0707.4470_F00171	25	0.438	`Q . \mathrm{push}\left(t_{\sigma}^{\text {next }}, \sigma \right) / /`	$Q . \mathrm{push}\left(t_{\sigma}^{\text {next }}, \sigma\right) / /$	Q.push$(t_\sigma^{\text{next}}, \sigma)$ // *Schedule σ for next update*
0707.4470_F00172	25	1.000	`x`	x	domly partitioned the x- and y-axes, so that each element has completely unique

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0707.4470_F00173	25	1.000	`y`	y	domly partitioned the x- and y-axes, so that each element has completely unique
0707.4470_F00174	27	0.998	`3+1`	$3+1$	so a 3+1 mesh (as in the previous subsections) is almost always adequate, and
0707.4470_F00175	29	0.919	`\mathscr{g}`	$\mathscr{g}$	Given a discrete 1-form A and dual source 3-form $\mathscr{J}$, define the discrete
0707.4470_F00176	29	1.000	`\mathrm{d} * \mathrm{~d} A=\mathscr{J}`	$\mathrm{d} * \mathrm{~d} A=\mathscr{J}$	Euler–Lagrange equations are therefore $\mathrm{d}{*}\mathrm{d}A = \mathscr{J}$. Defining the discrete 2-forms
0707.4470_F00177	29	—	`\square`	$\square$	—
0707.4470_F00178	29	1.000	`\theta_{\mathscr{L}}`	$\theta_{\mathscr{L}}$	to vanish at the boundary. As originally presented, the Cartan form $\theta_{\mathscr{L}}$ is an
0707.4470_F00179	29	1.000	`(n+1)`	$(n+1)$	$(n+1)$-form, where the n-dimensional boundary integral is then obtained by
0707.4470_F00180	29	0.976	`(n+2)`	$(n+2)$	contracting $\theta_{\mathscr{L}}$ with a variation. The multisymplectic $(n+2)$-form $\omega_{\mathscr{L}}$ is then
0707.4470_F00181	29	0.976	`\omega_{\mathscr{L}}`	$\omega_{\mathscr{L}}$	contracting $\theta_{\mathscr{L}}$ with a variation. The multisymplectic $(n+2)$-form $\omega_{\mathscr{L}}$ is then
0707.4470_F00182	29	1.000	`\omega_{\mathscr{L}}=-\mathbf{d} \theta_{\mathscr{L}}`	$\omega_{\mathscr{L}}=-\mathbf{d} \theta_{\mathscr{L}}$	given by $\omega_{\mathscr{L}} = -\mathbf{d}\theta_{\mathscr{L}}$. Contracting $\omega_{\mathscr{L}}$ with two arbitrary variations gives an
0707.4470_F00183	29	1.000	`\mathbf{d}^{2}=0`	$\mathbf{d}^{2}=0$	*multisymplectic form formula*, which results from the identity $\mathbf{d}^2 = 0$. In the
0707.4470_F00184	29	1.000	`n=0`	$n=0$	special case of mechanics, where $n=0$, the boundary consists of the initial and
0707.4470_F00185	29	1.000	`\omega_{L}`	ω_{L}	ω_L is preserved by the time flow.
0707.4470_F00186	30	1.000	`K \subset \mathscr{K}`	$K \subset \mathscr{K}$	*Proof.* Let $K \subset \mathscr{K}$ be an arbitrary subcomplex, and consider the discrete action
0707.4470_F00187	30	1.000	`S_{d}`	S_{d}	functional S_d restricted to K. Suppose now that we take a discrete variation α,
0707.4470_F00188	30	1.000	`\theta_{\mathscr{L}_{d}}`	$\theta_{\mathscr{L}_{d}}$	Then we can define the *Cartan form* $\theta_{\mathscr{L}_d}$ by
0707.4470_F00189	30	1.000	`\omega_{\mathscr{L}_{d}}=-\mathbf{d} \theta_{ \mathscr{L}_{d}}`	$\omega_{\mathscr{L}_{d}}=-\mathbf{d} \theta_{\mathscr{L}_{d}}$	Finally, defining the multisymplectic form $\omega_{\mathscr{L}_d} = -\mathbf{d}\theta_{\mathscr{L}_d}$, and using the fact that
0707.4470_F00190	30	1.000	`\mathbf{d}^{2} S_{d}=0`	$\mathbf{d}^{2} S_{d}=0$	$\mathbf{d}^2 S_d = 0$, we get the relation
0707.4470_F00191	30	1.000	`\alpha, \beta`	α, β	for all variations α, β; Equation 5.2 is a discrete version of the multisymplec-
0707.4470_F00192	31	0.998	`\mathscr{K}`	$\mathscr{K}$	here a 2-D asynchronous-time mesh $\mathscr{K}$, where the shaded region is an arbitrary
0707.4470_F00193	31	1.000	`\omega_{\mathscr{L}_{d}}`	$\omega_{\mathscr{L}_{d}}$	plectic form $\omega_{\mathscr{L}_d}$, the formula states that $\omega_{\mathscr{L}_d} \cdot \alpha \cdot \beta$ vanishes when integrated
0707.4470_F00194	31	1.000	`\omega_{\mathscr{L}_{d}} \cdot \alpha \cdot \beta`	$\omega_{\mathscr{L}_{d}} \cdot \alpha \cdot \beta$	plectic form $\omega_{\mathscr{L}_d}$, the formula states that $\omega_{\mathscr{L}_d} \cdot \alpha \cdot \beta$ vanishes when integrated
0707.4470_F00195	31	1.000	`\rho=\nabla \cdot \mathbf{D}`	$\rho=\nabla \cdot \mathbf{D}$	turns out to be the charge density $\rho = \nabla \cdot \mathbf{D}$, which justifies its elimination from
0707.4470_F00196	31	1.000	`f`	f	are invariant under gauge transformations of the form $A \mapsto A + \mathrm{d}f$, where f is any
0707.4470_F00197	31	1.000	`A_{t}=0`	$A_{t}=0$	Weyl gauge, so that the time component $A_t = 0$. Therefore, we can assume that
0707.4470_F00198	31	0.749	`A_{x}, A_{y}, A_{z}`	A_{x}, A_{y}, A_{z}	In fact, A_x, A_y, A_z are precisely the components of the familiar vector potential $\mathbf{A}$,
0707.4470_F00199	31	0.749	`\mathbf{A}`	$\mathbf{A}$	In fact, A_x, A_y, A_z are precisely the components of the familiar vector potential $\mathbf{A}$,
0707.4470_F00200	31	0.880	`A=\mathbf{A}^{b}`	$A=\mathbf{A}^{b}$	i.e., $A = \mathbf{A}^\flat$.
0707.4470_F00201	31	1.000	`\mathrm{d}_{t}`	d_{t}	nate, we can now define two new "partial exterior derivative" operators, d_t (time)
0707.4470_F00202	32	0.934	`\mathrm{d}_{s}`	d_{s}	and d_s (space), where $\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$. Since A contains no $\mathrm{d}t$ terms, $\mathrm{d}_s A$ is a 2-form
0707.4470_F00203	32	0.934	`\mathrm{d}=\mathrm{d}_{t}+\mathrm{d}_{s}`	$\mathrm{d}=\mathrm{d}_{t}+\mathrm{d}_{s}$	and d_s (space), where $\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$. Since A contains no $\mathrm{d}t$ terms, $\mathrm{d}_s A$ is a 2-form
0707.4470_F00204	32	0.934	`\mathrm{d} t`	$\mathrm{d} t$	and d_s (space), where $\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$. Since A contains no $\mathrm{d}t$ terms, $\mathrm{d}_s A$ is a 2-form
0707.4470_F00205	32	0.934	`\mathrm{d}_{s} A`	$\mathrm{d}_{s} A$	and d_s (space), where $\mathrm{d} = \mathrm{d}_t + \mathrm{d}_s$. Since A contains no $\mathrm{d}t$ terms, $\mathrm{d}_s A$ is a 2-form
0707.4470_F00206	32	1.000	`\mathrm{d}_{t} A`	$\mathrm{d}_{t} A$	containing only the space terms of $\mathrm{d}A$, while $\mathrm{d}_t A$ contains the terms involving
0707.4470_F00207	32	0.999	`\mathrm{d}_{s} D=\rho`	$\mathrm{d}_{s} D=\rho$	law as the sole Euler–Lagrange equation. The divergence constraint $\mathrm{d}_s D = \rho$,
0707.4470_F00208	32	1.000	`t_{f}`	t_{f}	requirement that variations α be fixed at the initial time t_0 and final time t_f. Then,
0707.4470_F00209	32	1.000	`\Sigma`	Σ	where Σ denotes a Cauchy surface of X, corresponding to the spatial domain. If
0707.4470_F00210	32	1.000	`\alpha=\mathrm{d}_{s} f`	$\alpha=\mathrm{d}_{s} f$	we vary along a gauge transformation $\alpha = \mathrm{d}_s f$, then this becomes
0707.4470_F00211	33	1.000	`\mathrm{d}_{s} D-\rho`	$\mathrm{d}_{s} D-\rho$	This indicates that $\mathrm{d}_s D - \rho$ is a conserved quantity, a momentum map, so if
0707.4470_F00212	33	0.986	`\nabla \cdot \mathbf{D}`	$\nabla \cdot \mathbf{D}$	density $\nabla \cdot \mathbf{D}$ is conserved, but more generally that the *change* in charge is related
0707.4470_F00213	36	1.000	`\mathbb{R}^{3}`	$\mathbb{R}^{3}$	Nédélec, J.-C. (1980), Mixed finite elements in $\mathbb{R}^3$. *Numer. Math.*, **35** (3), 315–341.